NR4A3 gapmers in extraskeletal myxoid chondrosarcoma: 87 of 190 pair a wild-type parent through the gap at 10 bp

Author. Tristan D. McRae

Independent researcher, unaffiliated. Correspondence: trimcrae@gmail.com ORCID: 0000-0002-1823-1451

Running title. Parent pairing in NR4A3 junction gapmers

Abstract

Extraskeletal myxoid chondrosarcoma (EMC) is an ultra-rare sarcoma usually defined by an in-frame partner-gene fusion to *NR4A3*. That junction is in no normal transcript, so an antisense gapmer could in principle cleave the fusion and spare its parents. This work is computational; every sequence named is a research reagent not for administration. Of 190 junction-spanning 16-mers tiled at 5-6-5 across the 38 in-frame junctions of five modelled partners, 87 let a mature wild-type parent transcript pair their whole catalytic gap over ten or more contiguous base pairs, 61 of those against wild-type *NR4A3* itself. Ten is a convention, not a measurement: exon-terminus chimeras of these transcripts meet the same screen at 40.6% against the panel's 45.8%, so most of that liability follows from joining exon termini, not this disease. Two reagents are named at the most reported breakpoints, 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6 to *NR4A3* exon 3, both at the panel's top gap-level margin of three, longest wild-type parent gap duplexes eight and nine base pairs. Five test articles are named; the two fusion-positive EMC cell models are reported at an *NR4A3* exon-2 acceptor and match different designs. The design pipeline is released.

Keywords

antisense oligonucleotide; gapmer; RNase-H1; fusion transcript; NR4A3; extraskeletal myxoid chondrosarcoma; off-target screening

1 · Background

EMC is defined in the large majority of cases by an in-frame fusion of *EWSR1* to *NR4A3*,[1] with *TAF15* a substantial minority and *TCF12* and *TFG* rare.[2] The disease responds poorly to conventional cytotoxic chemotherapy,[3] though responses do occur,[4] and a tyrosine-kinase inhibitor trialled in it gave disease control more often than response — a reading composed from the review's response categories[3] rather than from a figure stated in the trial report.[5]

The fusion junction is the one feature of an EMC tumour that exists at the RNA level and in no normal cell. An antisense gapmer tiled across it recruits RNase-H1 to cleave the transcript it pairs, at the six-nucleotide DNA gap at the centre of a 5-6-5 architecture. Junction-directed nucleic-acid agents are a thirty-five-year lineage, reported against at least six fusion oncogenes: two as antisense oligonucleotides, the rest as RNA-interference agents, one of those from a lentiviral vector rather than an administered oligonucleotide.[6,7,8,9,10,11] No such design is reported for any *NR4A3* fusion in the literature retrieved here, and that absence is why this work exists.

What a junction design must survive follows from its construction. Both halves are parent-gene sequence, so each parent matches roughly half the oligonucleotide — far outside the mismatch budget of a conventional off-target search, which therefore never returns a parent as a near-match. RNase-H1 does not require the whole duplex, only that the gap be paired — a premise adopted here rather than established, and one whose length requirement is stated in a different unit from the criterion this paper screens on. The requirement is reported as a DNA gap of at least six nucleotides, with seven to ten the working range; the screen below counts a liability only at ten contiguous base pairs of duplex through that gap, a length of hybrid rather than a count of gap nucleotides. Whether a wild-type parent pairs the catalytic gap contiguously is therefore a separate question from overall similarity, and it is the one this work puts to all 190 designs before recommending any.

2 · The reagents

The two reagents named for synthesis are the best available designs at the two junctions with a published exon-resolved breakpoint and the highest reported prevalence: 5′-GGGCATATCATCAAAC-3′ at *EWSR1* exon 12 joined to *NR4A3* exon 3, and 5′-GGGCATATCTTGTGTG-3′ at *TAF15* exon 6 joined to *NR4A3* exon 3 (Table 1). Exon numbers throughout are transcript exon indices counted from the transcript 5′ end, including non-coding exons; an acceptor exon number read under the coding-exon convention instead selects a different reagent. Both hold the panel's top gap-level margin of three: three junction-unique bases inside the catalytic gap on the shorter side of the breakpoint, and neither pairs a wild-type parent through the gap at §3's ten-base-pair criterion.

Both sit close to it. The *EWSR1* reagent's longest wild-type parent duplex through the whole gap is eight base pairs and the *TAF15* reagent's is nine, both against wild-type *TFG*, so the cut decides how they read: at eight both fall inside the class this work marks as not to be ordered, at nine the *TAF15* reagent alone does, and only at ten does neither. Both also pair a wild-type parent through part of the gap at the *NR4A3* exon-2/exon-3 seam their acceptor halves share, neither in full; those partial duplexes are not counted here and have not been measured. Consecutive registers of one seam differ by a single-base slide and can carry opposite verdicts (Table 2).

Predicted transcriptome load separates the two: 123 gap-paired sense-strand near-matches for the *EWSR1* reagent at a deeper search ceiling than the default, against eight for the *TAF15* one. Most of the 123 are predicted transcript models rather than curated records, so the count is a search result over a database that includes predictions, not a census of expressed transcripts. The *EWSR1* reagent also carries a sense-strand near-match in wild-type *TAF15* precursor RNA at two mismatches, one inside the catalytic gap, spanning an intron-exon boundary: the cost of the same ten shared donor bases that let one oligonucleotide span the *EWSR1*, *TAF15* and *FUS* breakpoints at once (Figure 1). The *TAF15* reagent carries no sense-strand precursor site. Both loads are predictions from sequence search rather than measured activity.

Both reagents are phosphorothioate throughout, with wings of five contiguous β-D-oxy-locked residues — a high locked content against the two to four per wing taken here as usual, so these are not conventional locked-nucleic-acid reagents and their matched-duplex melting temperatures are correspondingly high. Both begin 5′-GGG, a contiguous locked G-tract. High affinity is taken to carry a risk of sequence-dependent hepatotoxicity; that is a premise adopted here rather than a retrieved finding, and nothing here measures it.

Discounted by the breakpoint distribution of an 18-case series,[12] the two junctions account for 68.4% of molecularly confirmed cases in a 58-case cohort.[13] That prices which published junctions the two reagents address; it is not a coverage measurement, no patient having been screened with either sequence. The range 39.9% to 82.8% quoted with it is not a confidence interval and carries no nominal level: two of its four inputs do not vary, and it assumes a breakpoint distribution reported in one cohort transfers to a second collected twenty-one years later. The *TAF15* arm is priced at three of three reported breakpoints, an upper bound rather than an estimate.

That figure is not a ceiling: a third design at *EWSR1* exon 13 joined to *NR4A3* exon 3, already tiled and screened at the same top margin, would take the figure above to 79.0%. It is not named for synthesis, on grounds the extended report sets out.

3 · Selection from a panel of 190 designs

The two reagents above are what survived a screen applied uniformly to the whole panel. Across the 38 in-frame junctions of five modelled partners, 190 junction-spanning designs were tiled and put through five specificity screens. Of those, 87 let one of six mature wild-type parent transcripts pair their entire catalytic gap over a contiguous duplex of at least ten base pairs, and 61 of the 87 do so against wild-type *NR4A3*; 85 are paired by one of the design's own two parent genes. A second class is invisible to any screen over mature transcripts: 19 designs carry a sense-strand near-match in parent precursor RNA pairing the gap in full. As a union rather than a sum the two screens condemn 93 of the 190. A design whose gap carries a mismatch is scored zero rather than short, so the 87 bound the fully-paired class, not the whole parent liability.

Lengthening the catalytic gap does not remove this liability, for a reason that is arithmetic rather than empirical: every base inside the gap comes from the donor or the acceptor exon, so a longer gap buys margin only by conceding parent-paired gap DNA at the design's own seam. Screened across three geometries the liable count does not fall, and at 5-10-5 the criterion is not independent of the geometry at all; the extended report gives the series.

Three designs clear every screen applied here, none at a junction any patient is reported to carry, which makes them mechanism controls rather than candidates. Selecting within each junction rather than across the panel is what makes the two named reagents available: 35 of the 38 junctions have a design that clears the parent screen, and all five junctions with a published exon-resolved breakpoint have one. Both readings are properties of the cut. At nine base pairs 31 of the 38 still clear and three of the five published ones do; at eight, 23 and two; at seven, 9 and none; at six, 6 and none. Those cuts are whole-duplex run lengths, not the enzyme's own unit, so no cut on this ladder is the enzymology restated, and the availability the two named reagents rest on fails at a run cut of nine, where the *TAF15* junction's best design is itself liable. Read as design counts the loose cuts condemn almost everything: 175 of 190 at seven and 181 at six.

Two bounds on the cleanliness claim: seven of the 190 screens never returned, and the alignment screen censors the rest, leaving 47 of 183 assessable at all, so a count of clean designs is a floor over that subset and not a total over the panel; and most designs clean at the default search ceiling are not clean at a deeper one.

The ten-base-pair criterion is adopted rather than measured, and the comparison against null models does not resolve an excess specific to this disease. Mononucleotide scrambles — the weakest of the ten nulls screened, and not the dinucleotide-preserving control §5 prescribes — meet the parent screen at 6.2% against 45.8% for the panel, but chimeras built at real exon termini of the same two transcripts, at junctions almost never reported in a patient, meet it at 40.6%. The adopted cut does not escape that comparison: at ten the strongest null's 40.6% falls inside the panel's own 95% interval on 45.8%, as at every cut from seven to thirteen but eleven, and the strongest null reaches 91.4% at seven against the panel's 92.1%. The comparison is narrower than it reads, the panel arm being itself mostly unreported junctions — the property the chimeric null is discounted for. Most of the liability is therefore what joining two exon termini of these genes gives, and across cuts of six to thirteen base pairs the excess over the strongest null changes sign four times.

4 · Test articles

Five test articles bear on the junctions this panel designs against, and they divide into two sources with opposite limits. Every design in the panel joins its donor to *NR4A3* exon 3; the two cell models are reported at exon 2, so they carry a panel junction only under the exon-3 reading their acceptor index leaves open.

Three are engineered constructs from a published functional study,[14] whose exon spans that paper states verbatim; two of them, E-N and T-N*, carry exactly the junctions the reagents above span, so both named reagents have a stated test article. Rebuilding them is the faster route, but a complementary DNA over-expressed in a heterologous background speaks to junction-selective knockdown of the intended transcript, not to activity at an endogenous locus.

The other two are patient-derived, identity-clean models reported with two EMC tumours,[15] USZ20-EMC1 (RRID:CVCL_C6MX) and USZ22-EMC2 (RRID:CVCL_C6MY) — the only source of a fusion-positive EMC cell identified here. They are available on request with no repository deposit, and are slow, at reported doubling times of five to six days. Their fusions are reported as *EWSR1* exon 13 and *TAF15* exon 6 joined to *NR4A3* exon 2 rather than exon 3, but that acceptor index is not settled: the report carries no sequenced exon-exon boundary, no transcript accession and no junction sequence, and this work's own withdrawn version arose from an error of exactly this class. Reagents exist at both acceptors, and in neither case is it the same molecule as the reagent named above: 5′-AGTGGGCTCTCCACGG-3′ at *EWSR1* exon 13 to exon 2 and 5′-AGTGGGCTCTTGTGTG-3′ at *TAF15* exon 6 to exon 2, both at the panel's top margin. Neither reaches the ten-base-pair criterion, and their longest wild-type parent duplexes through the whole gap are eight base pairs against wild-type *EWSR1* and nine against wild-type *NR4A3* — the second against the acceptor parent on which §5's selectivity ratio is defined, a closer call than either exon-3 reagent presents. A reagent selected for one acceptor is not valid for the other.

One requirement is upstream of all of them: the breakpoint of the test article must be established at nucleotide resolution by RNA sequencing before any oligonucleotide is ordered, every design here being specific to the exon pair it was tiled at. Routine diagnosis does not supply the seam, since break-apart *NR4A3* fluorescence in situ hybridisation detects a rearrangement irrespective of partner.[3]

5 · The falsification experiment

An isogenic fusion-positive against fusion-negative comparison has been run in an analogous fusion sarcoma, against *NAB2::STAT6* in solitary fibrous tumour.[16] The extended report specifies it for these reagents, with three controls a knockdown assay alone cannot distinguish. One belongs here because it constrains what may be ordered: the dinucleotide-preserving scramble must itself pass the mature-parent screen before synthesis, because 10.0% of dinucleotide-preserving scrambles pair a parent's whole catalytic gap at the ten-base-pair criterion and 3.9% do so against wild-type *NR4A3*.

Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose-response in the same wells, at a cut of 5.0 adopted as a convention. At a replicate standard deviation of 0.35 on the natural-log scale, six independent biological replicates give about 80% power to falsify a true selectivity of 3 and three give about 30%. Above a realised standard deviation of about 0.65 at three replicates — 1.53 at six, 2.25 at ten — no observed ratio at or above one can put the upper limit of a two-sided 95% interval below 5, so the test can fail only on an anti-selective reading. Those figures are computed on that interval; a normal-approximation interval would move them. Such a test is void, and voidness is a property of the realised variance rather than of the design, so a pilot-based gate on the population variance bounds it from one side only: where a one-sided 95% chi-square bound on the pilot's standard deviation lies at or above the void figure for the count proposed, the decision is a larger replicate count, or no falsification test at all, and never three. The threshold is defined on the acceptor parent alone, so a reagent can clear it while pairing a donor transcript through its whole gap.

6 · Beyond the panel

The panel is bounded by what has been sequenced rather than by what can be designed, and the procedure that produced the 190 designs is released unchanged with the artefacts. It is not a turnkey service for an arbitrary breakpoint: the builder's published-breakpoint list is a waiver list rather than a gate, admitting seams that would otherwise be refused for a non-coding acceptor or an out-of-frame join. A seam needing such a waiver requires the list to be extended first; a seam needing none is emitted without any check that a patient has been reported to carry it.

A design is certifiable where all five screens could be run on it and it cleared all five; one that a screen cannot address is uncertifiable whatever the others return. The reagent for the third engineered construct of §4 is uncertifiable on that definition, its seam lying outside the compartment three of the five screens reach. The two that reach it — the precursor-RNA and genome arms — agree design by design, clearing the seam's top-margin design and condemning a lower-margin one at a wild-type *NR4A3* site each records as pairing the whole catalytic gap; two screens agreeing is not the clearance the other three would supply.

7 · Discussion

Designability is not the constraint: junction-spanning designs exist at every in-frame *NR4A3* fusion junction modelled here, and 35 of the 38 have one clearing the parent screen.

The constraint is discrimination between the fusion and its parents, and it is not resolved here. A junction design's most plausible wild-type liability is its own parent, in the mature transcript or across a splice junction in precursor RNA. Both are searched before any molecule exists, but not on comparable terms: the mature-transcript screen condemns on a ten-base-pair duplex through the gap and the precursor arm on a hit at up to two mismatches with the gap fully paired, so neither restates the other, and their counts may not be added because a design condemned in both is one design. A third compartment is searched only in the extended report: the patient's own un-rearranged *NR4A3* allele, at a two-mismatch ceiling that bounds its class from below. No design this panel selects is condemned by it, but it excludes two registers of §4's *EWSR1* exon 13 to *NR4A3* exon-2 seam while clearing that reagent — the register hazard of §2 arriving from a compartment the panel's selection never has to consider.

The four junction-specificity reports cited here[8,9,10,17] were all made on molecules already synthesised; no survey of design pipelines was performed, so the screen-before-synthesis claim is about this literature as retrieved and not a priority claim. Whether sparing wild-type *NR4A3* is worth a specificity cost is itself unsettled. *NR4A1* and *NR4A3* are functionally redundant where tested,[18,19] which cuts against the premise; against that, reduced *NR4A1*/*NR4A3* dosage is consequential in mice,[20] and the family is not uniform in direction, *NR4A3* aggravating murine atherosclerotic lesions where its paralogues attenuate them.[21] The evidence is not decisive either way.

Three limits bound what any test of these reagents could show. All five screens address hybridisation rather than cleavage, and none establishes that a predicted duplex forms or is cut. The method-level novelty is nil, junction-directed oligonucleotides being long established; what is new is the indication and the screen applied before synthesis. And systemic delivery to a solid tumour remains unsolved, the gate this modality faces after any result reported here. Every source of a test article named here ends at someone culturing cells.

8 · Methods

All analyses are computational and use public data; no laboratory work was performed. Full parameters, the complete bounds on each claim and the per-design tables are in the extended report named under Data availability.

Canonical transcripts for the five partner genes and for *NR4A3* were obtained from Ensembl.[22] Junction-spanning 16-mer gapmers were tiled in a 5-6-5 β-D-oxy-locked-nucleic-acid/DNA/β-D-oxy-locked-nucleic-acid geometry,[23] one design per register at which the breakpoint falls inside the six-nucleotide DNA gap, which admits five per junction. That gap is the shortest length the cited source credits rather than its preferred one, and six is used because the genome-wide arm is not available above a 16-mer. A design's gap-level margin is the count of junction-unique bases inside the gap on the shorter side of the breakpoint.

Five specificity screens were applied: an alignment screen against human RefSeq RNA, classifying each near-match by whether the catalytic gap is paired; an exhaustive transcript scan complete for substitutions within a one-mismatch budget; a screen of the parents' unspliced sequence; a mature-parent screen recording the longest contiguous duplex any of six wild-type parent transcripts forms through the catalytic gap; and a genome-wide screen over every position of GRCh38. A near-match is a transcript window pairing a design at 14 or more of its 16 positions. A design is liable where a wild-type parent pairs its whole catalytic gap over a contiguous run of ten base pairs or more, ten being adopted rather than measured. Null ensembles were built as scrambles of each design and as chimeras joining the same two parent transcripts at real exon termini, screened identically.

Tables

Every cell below is a column of `fusion-junction-aso-sequences.csv`, the canonical machine-readable record, except the test-article column of Table 1, which is a literature fact and carries its source in the caption. Every reagent named here is a 5-6-5 phosphorothioate gapmer. An oligonucleotide should be ordered from that file rather than transcribed from this page.

Table 1. The two reagents named for synthesis, with their parent-duplex label and their test article. The parent-duplex column is the longest contiguous duplex a mature wild-type parent forms through the catalytic gap; neither reagent reaches the ten-base-pair criterion, so the length is printed rather than a pass mark. Test articles are the engineered constructs of Brenca et al. (PMID: 31020999); the two patient-derived models of Bangerter et al. (PMID: 36316541) are REPORTED at an NR4A3 exon-2 acceptor and match different designs, not these two. Nothing here has been synthesised or tested, and no sequence may be administered to any person or animal.

Table 1. The two reagents named for synthesis, with their parent-duplex label and their test article

seam	reagent	margin	WT gap duplex (bp)	test article
EWSR1 e12::*NR4A3* e3	5′-GGGCATATCATCAAAC-3′	3	8 bp, wild-type *TFG*	E-N, engineered construct
TAF15 e6::*NR4A3* e3	5′-GGGCATATCTTGTGTG-3′	3	9 bp, wild-type *TFG*	T-N*, engineered construct

Table 2. Two near-identical designs at one seam, one orderable and one not. Each pair is two consecutive registers of one seam differing by a single-base slide, and the two members carry opposite verdicts: the condemned member pairs its whole catalytic gap against a wild-type parent at the ten-base-pair criterion and the orderable member does not. The condemned class reaches the reagents this paper names for synthesis: 5′-AGGGCATATCTTGTGT-3′ is one single-base slide from 5′-GGGCATATCTTGTGTG-3′ and pairs 11 bp of wild-type *NR4A3* through its whole catalytic gap. Neither may be substituted for the other, and neither is named for synthesis. The pairing is read from the canonical file's own cross-reference column.

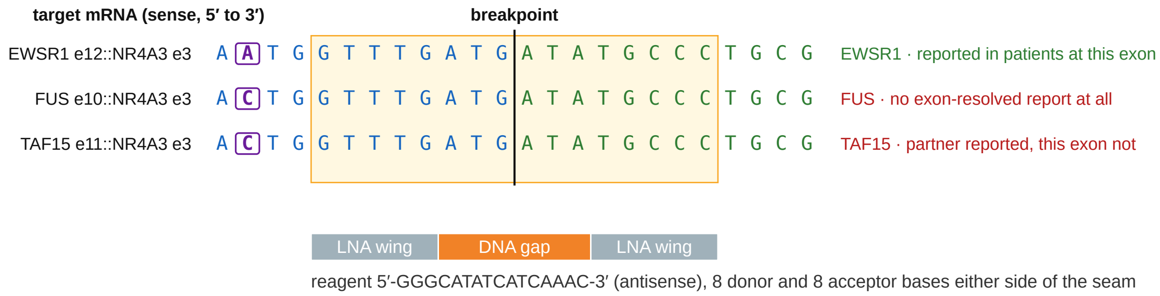
Table 2. Two near-identical designs at one seam, one orderable and one not

seam	design	verdict	margin	WT gap duplex (bp)
EWSR1 e12::*NR4A3* e3	5′-CAGGGCATATCATCAA-3′	DO NOT ORDER	1	11 bp, wild-type *NR4A3*
EWSR1 e12::*NR4A3* e3	5′-AGGGCATATCATCAAA-3′	orderable (a single-base slide)	2	7 bp, wild-type *TFG*

Figure legends

One 16-mer spans three partners' breakpoints — and only one of the three is a junction any patient is reported to carry

The three FET-family donors are identical over the ten nucleotides before the breakpoint, and the acceptor exon is the same in all three. Every row is TARGET mRNA, sense; the reagent is the reverse complement of the shaded window.



Blue, donor exon; green, NR4A3 acceptor exon; boxed and purple, the one position at which the three donors differ. Shaded box, the window this reagent targets.

The reagent is 5′-GGGCATATCATCAAAC-3′, the reverse complement of the shaded window (5′-GTTTGATGATATGCCC-3′). Transcribing the shaded letters orders the sense strand, which is a different molecule with no antisense activity.

Research use only, not for administration. The bases alone, ordered as unmodified DNA, are a different molecule; order from fusion-junction-aso-sequences.csv.

Figure 1. One 16-mer spans three partners' breakpoints, and only one of the three is a junction any patient is reported to carry. The junction windows of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, aligned at the breakpoint, each row carrying its own reporting status.

Author Contributions

T.D.M. is the sole author, and is responsible for the conception and design of the study, the analysis code, the screens and their interpretation, and the drafting and revision of this manuscript.

Statements and Declarations

Research use only, and not for administration to any person or animal. Every sequence here is a research reagent for laboratory investigation only, and none has been synthesised or tested. Order from the canonical record, `fusion-junction-aso-sequences.csv`, which specifies the sequence, every locked residue and the backbone, and not until the breakpoint has been established at nucleotide resolution by RNA sequencing.

Ethical considerations. Not applicable. No human subjects, human material or animals were involved, and no ethics approval was required.

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Consent for publication. Not applicable. The manuscript contains no data from any individual person.

Declaration of conflicting interest. The author declares no financial competing interests: he holds no patent, patent application, equity or consultancy relating to any sequence or method described here. A non-financial interest is disclosed to the editor in the accompanying cover letter.

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Use of artificial intelligence. A large language model (Claude, Anthropic) was used throughout this work: to write and review the analysis code, to run the screens, to retrieve and check literature, and to draft and revise this manuscript. The author directed all work reported here and is responsible for its content.

Data availability. All code, graded artefacts and per-design tables are deposited under doi:10.5281/zenodo.22061075. The extended report, carrying every screen's parameters and the complete bounds on each claim, is `fusion-junction-aso-research-article.pdf` inside that deposit; it is not posted as a preprint, so the archived copy is the citable one. An earlier version of these analyses placed the acceptor junction incorrectly through a coding-versus-transcript exon indexing error and was withdrawn in full; the panels were rebuilt and verified, and the complete correction record is released with the archive.

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Numbered contiguously by order of first citation in this article. A reference cited in both this article and the extended report may therefore carry a different number in each, and a superscript here resolves only against this list. Metadata is read from retrieved bibliographic records.

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